Hillcrest (Wine Glass) pass ranges SUGGESTION

Hillcrest AL (Wine Glass), 540235, the repeater on 151.500 MHz at Logan · the station register as of 4 November 2026 · what Flood-Net's receiver heard, 5 October to 4 November 2026 · written 4 November 2026.

A **pass range** is a block of ALERT addresses a repeater is set to relay. Hillcrest relays every reading it hears whose address is in one of its ranges, and sends it out again on the channel it heard it on, 151.500.

The answer

Cut Hillcrest from 621 addresses to 167. Keep Logan's stations and the few stations no other repeater on 151.500 carries. Leave the Ipswich, Lockyer, Moreton Bay and Brisbane blocks, and the Scenic Rim block, to the repeaters that already carry them.

Now	Suggested	What the suggested range carries
1900–2200	**2051–2053**	Springfield Lakes AL (Ipswich City Council)
	2065–2067	Grandchester AL (Bureau), kept to be safe, see *Watch*
	2101–2102	Jingle Downs AL
	2179–2181	Rosewood Detention Basin AL (Ipswich City Council), kept to be safe
4151–4199	**4151–4199** (no change)	Logan's block: Schmidts Rd, Bayes Rd, Lower and Upper Quinzeh, Waller Rd, Henderson Ck, and room to grow
4800–4899	**4840–4845**	Springfield Lakes HL Dam and LL Dam
5304–5399	**5308–5312**	Romani, Maclean Bridge (Logan City Council)
	5356–5367	Mt Alford, Mt Sugarloaf, Wyaralong Dam (Nindooinbah and Bromelton Dams come along, at no cost)
	5391–5399	Darlington School, Innisplain
6200–6274	**6200–6274** (no change)	Hillcrest's own rain and battery, and Logan's block: 18 stations from Oxley Ck to Beenleigh

An ERT-A2 takes all ten. It has four address lists: one Pass list and up to three Reject lists. Each list holds up to 32 entries, and an entry is one address or a range (ELPRO, ERT-A2 User Manual, Sept 2025, p. 27). Only one list can pass, so put all ten in that list: *Communication*, then *Address List* → List 1, *Address List Status* → Enable, *Address List Action* → Pass, then *Address List Entries*. Delete what is there, add the ten, and save in *Show/Save Configuration*. Grandchester (2065–2067) and Rosewood Detention Basin (2179–2181) are kept only to be cautious. They can come out later if the week's watch shows Constitution Hill carries them.

What it does:

- **Hillcrest sends about 7 in 10 fewer relays.** 15,469 of the 22,333 bursts heard on its carrier from 22 October to 4 November were from addresses the suggested ranges leave out.
- **No station is left without a repeater.** Every station on file that Hillcrest carries today is still carried by at least one repeater afterwards. That includes all 24 Logan City Council stations, all 16 Ipswich City Council stations, all 11 City of Moreton Bay stations, all 7 Scenic Rim Regional Council stations and all 6 Brisbane City Council stations. 39 stations stay on Hillcrest and 113 go to repeaters that already carry them.

- **Its overlap with Constitution Hill falls from 319 addresses on file to 87.** That was the busiest pair of repeaters in the delay plan (repeater-delays.md).

Why it is too open

1900–2200 and 4800–4899 copy Constitution Hill's ranges exactly, and Constitution Hill is itself a base. Together they cover 94 stations in this area, from Beachmere and Dayboro, 60 km north, to Buaraba and Grandchester, 56–70 km west. For 86 of them, Constitution Hill's modelled fade margin (how much signal is to spare on the path) is better than Hillcrest's, by up to 49 dB. Of the other eight, four stay on Hillcrest (Grandchester, Jingle Downs and the two Springfield Lakes gauges). The other four are Calamvale, Forestdale, Greenbank and Mt Marrow, where Constitution Hill still has 16–34 dB. Whatever Constitution Hill hears has already reached the Bureau, so Hillcrest's copy adds nothing.

Hillcrest re-sends Constitution Hill's relays. Timing shows it. Each relay comes a fixed time after the burst it repeats: about half a second here, one hop. Where Flood-Net's receiver heard an earlier copy of the same reading:

- For **Logan's stations**, Hillcrest's copy came about 0.5 s after it (26 of 28 times). That is one hop: Hillcrest heard the station itself.
- For the **Logan–Albert stations in 5308–5399**, the same: 107 of 111 one hop.
- For the **Ipswich, Lockyer and 4800–4899 stations**, more than 8 in 10 came 0.9 s or more after an earlier copy, most of them 1 to 1.6 s. So Hillcrest was the second or third transmitter of that reading. John Bray Park is plain: 240 of 253 of Hillcrest's copies came 0.5–2.8 s after another copy, most of them 1.0–1.4 s.

So on 151.500, one reading from Ipswich often goes out three times or more: the station's own burst, Constitution Hill's relay and Hillcrest's. That airtime is shared with every station and repeater on the channel. It matters most in a storm, when gauges send fast and bursts collide. Each needless relay is also another chance for the small damage Hillcrest does (1.6% of the relays checked, investigation 03).

5313–5399 is already carried onto 152.400. Laheys Lookout, O'Reillys, Wilsons Peak and Kooralbyn carry those stations, and Flood-Net's receiver heard them there hundreds of times in 20 days: for example Kilmoylar Rd 865 times, Yarrahappini 670, Bromfleet 588 and Nindooinbah Dam 551. Hillcrest's copies of most of them were few, or second-hand.

Its ranges also hold 67 far-off stations that use the same addresses on other networks: 54 in 4800–4899 (Gippsland, Alice Springs and Bundaberg) and 13 in 5304–5399 (South Australia). Hillcrest can never hear them. The ranges were never cut to fit this area.

Nobody is orphaned — the council stations

Council	Stations in Hillcrest's ranges	Stay on Hillcrest	The others are carried by
Logan City Council	24	19	Laheys Lookout, heard on 152.400 460–865 times each in 20 days (Bromfleet, Yarrahappini, Tamborine, Flagstone Ck, Kilmoylar Rd)
Ipswich City Council	16	2	Constitution Hill. Modelled margin from Hillcrest is worse for all 14
City of Moreton Bay	11	0	Constitution Hill, and Round Mt Res for Beachmere and Lipscombe Rd
Scenic Rim Regional Council	7	0	Laheys Lookout, O'Reillys and Wilsons Peak onto 152.400, the channel the SRRC base listens on. Hillcrest never reached that base: it sends on 151.500
Brisbane City Council	6	0	Constitution Hill, 34–40 dB. Hillcrest relayed these six 11 times between them from 22 October to 4 November

Why Logan's stations stay. Hillcrest hears them first-hand. Its modelled margins are 8 to 54 dB, apart from Wolffdene and Bahrs Scrub, which are weak on every path. Constitution Hill's are −7 to 24 dB. The register lists Mt Tamborine as a second path for 4151–4199 and 6200–6274. But in 20 days of listening to 152.400, Flood-Net's receiver heard **none** of those addresses there, while Mt Tamborine's Gold Coast block came through more than 13,000 times. So for Logan, Hillcrest is the main path and Constitution Hill the only backup. Keep both.

Why the five stations in 5356–5399 stay. Mt Alford, Mt Sugarloaf, Wyaralong Dam, Darlington School and Innisplain are in no other 151.500 repeater's ranges. Constitution Hill No 2 lists them but listens on 152.400. Mt Alford sends on 151.500 and Hillcrest hears it first-hand (98 of 103 one hop). The rest are heard too seldom to be sure, and keeping them costs Hillcrest almost nothing.

Watch for a week after

1. **Check in the Bureau's own data that each station below still arrives.** Each one sends a report every three hours, rain or not, so a gap shows within a day. If one goes quiet, add its range back to Hillcrest. These are the stations taken off that have a weak modelled margin to Constitution Hill:
   - **Meadow Flat Rd Bridge** 4837–4839, **Marburg AL** 2086–2088, **Limestone Park** 4828–4830, **South Ripley (Wards Rd)** 4813–4815 (all Ipswich City Council); **John Bray Park** 1956–1958, **Samford Valley** 1962–1964, **Lipscombe Rd** 1979–1981 (City of Moreton Bay); **Greys Plains Rd** 2077–2079, **Goolman** 4818–4819, **Upper Western Ck** 4816–4817, **Lower Mt Walker** 2030–2032, **Peak Crossing** 2062–2064, **Tallegalla** 2068–2069. In every case Hillcrest's own margin is worse, and where the timing shows anything, Hillcrest's copy came second.
2. **Flood-Net's own receiver will hear less, and that is expected.** It hears about 48 of the stations being taken off mostly through Hillcrest (Ipswich AL, Moggill, Brisbane Rd, Karalee, Beachmere and others). Station Health may call them quiet. That is our receiver losing a copy, not the network losing the station. Constitution Hill's copies may come through clearer once Hillcrest stops landing on top of them.
3. **Put the new ranges on the register once they are set.** Use the Stations tab: Hillcrest's card, then its pass ranges. Then the Pass Ranges and Repeaters tabs and the delay plan follow the real setting.

Also found

- **Mt Tamborine's register ranges don't match the air.** Its 4151–4199 and 6200–6274 are never heard on 152.400. Either it isn't set to relay them, or it can't hear Logan. Worth checking at its next visit, and correcting on the register either way.
- **Something not on the register sends 4846–4848 and 4856.** It was heard 2,672 times on 151.500 in a month, on a steady carrier. The register has those addresses only on stations in Victoria. It is probably a station in this area that's missing from the register. Constitution Hill still carries 4800–4899, so this change doesn't cut it off, but it should be found and recorded.
- **Hillcrest and Constitution Hill still share 87 addresses** (Logan's blocks and a few others), so they still need different delays. The ⏳ Repeater Delays tab plans again once the register holds the new ranges.

How this was worked out

- **Ranges and stations:** `stations.json`. A station counts as carried if one of its ALERT addresses is in a repeater's pass ranges, as the Pass Ranges tab works it out. Owners come from the Bureau's Service Level Specification.
- **Margins:** the fade margins saved in Flood-Net (`meganet.link_fade_margin`, the field model), station to repeater, one way. Good is 15 dB or more, OK is 6–15 dB.
- **What was heard:** Flood-Net's receiver, legacy ALERT. On 151.500, 5 October to 4 November. On 152.400, 5 to 24 October, when it was listening there. Hillcrest's carrier is +120 to +240 Hz, as investigation 03 found; offsets were measured from 22 October.
- **Timing:** for each burst on Hillcrest's carrier, the earliest copy of the same address and value on another carrier in the five seconds before it.
- **Limits:** the receiver is one site and is not a Bureau base. A margin is a model, not a measurement. So the watch list above is part of the change, not an extra.